

Name of Institute: Indus Institute of Technology & Engineering

Name of Faculty: Prof. Ruchi Patel

Course code: CE0216

Course name: Programming for Problem Solving

Pre-requisites: Students must have a basic understanding of Computer Programming Terminology and Mathematics.

Credit points: 4

Offered Semester: II – All Branches

Course Coordinator

Full name: Prof. Ruchi Patel

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Consultation times: 3.00 P.M. to 5.00 P.M. (Monday to Friday)

Course Lecturer

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Full name: Prof. Zalak Vyas

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Consultation times: 3.00 P.M. to 5.00 P.M. (Monday to Friday)

Full name: Dr. Kaushal Jani

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Consultation times: 3.00 P.M. to 5.00 P.M. (Monday to Friday)

Full name: Prof. Hiren Mer

Department with sitting location: Bhawar Building, 4th Floor

Email: hirenmer.ce@indusuni.ac.in

Consultation times: 3.00 P.M. to 5.00 P.M. (Monday to Friday)

Students will be contacted throughout the session via mail with important information relating to this course.

Course Objectives

- 1) To familiarize the student with basic concepts of computer programming and developer tools.

- 2) To describe the parts of the computer system.
- 3) To describe functioning of computer components.
- 4) To describe the process of problem-solving using a computer.
- 5) To describe the design of an algorithmic solution for a given problem.
- 6) To describe the writing method for a maintainable 'C' program for a given algorithm.
- 7) To describe the importance of 'C' program for simple applications of real-life using structures and files.
- 8) The students will be able to enhance their analyzing and problem-solving skills and use the same for writing programs in 'C'.

Course Outcomes (CO)

By participating in and understanding all facets of this course a student will be able to:

- i. **Gain** a broad perspective about the usage of computers in the engineering industry.
- ii. **Develop** basic understanding of computer programs, concept of algorithm and algorithmic thinking.
- iii. **Develop** the ability to analyze a problem and write an algorithm and program to solve it.
- iv. **Develop** logical thinking ability and apply basic programming principles using 'C' language.
- v. **Development** of different 'C' programming features and structures in order to solve different problems considered as an initial step towards software development.
- vi. **Apply** fundamental principles of problem solving in software engineering through various programming languages.

Course Outline

UNIT-I	[12 hours]
Introduction to Programming	
What is programming? Problem solving methods with Examples-Algorithm and Flowchart, Types of Programming languages, Characteristics of higher level language, Some Programming languages	
Introduction to 'C'	
Introduction, Importance of C, Sample C programs, Basic structure of C programs, Programming style, executing a C program.	
Introduction, Character Set, C tokens, Keywords and Identifiers, Constants, Variables, Data types, Declaration of Variables, Defining symbolic constants	
Operators and Expression	
Introduction, Arithmetic of Operators, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operators, Bitwise Operators, Special Operators, Arithmetic Expressions, Evaluation of expressions, Precedence of arithmetic operators, Type conversions in expressions, Mathematical function.	
UNIT-II	[12 hours]
Decision Making Statements	

Introduction, Decision making with IF statement, Simple IF statement, the IF ELSE statement, Nesting of IF ... ELSE statements, The ELSE IF ladder, The switch statement, the ternary (? :) Operator, the GOTO statement.	
Looping	
WHILE statement, the DO statement, The FOR statement, Jumps in loops Break and continue.	
Array & Handling of Character strings	
Introduction, One-dimensional arrays, Two-dimensional arrays, Initialization of two-dimensional arrays, Concept of Multidimensional arrays	
UNIT-III	[12 hours]
Handling of Character String	
Introduction, Declaring and initializing string variables, reading string from terminal, writing string to screen, Arithmetic operations on characters, Putting string together, String Operations: String Copy, String Compare, String Concatenation and String Length, String Handling functions, Table of strings.	
User-Defined Functions	
Introduction, need for user-defined functions, return values and their types, calling a function, category of functions, no arguments and no return values, Arguments with return values, Handling of non-integer functions, Nesting of functions, Recursion, Functions with arrays, The scope and Lifetime of variables in functions.	
UNIT-IV [12 hours]	[12 hours]
Pointers	
Introduction, understanding pointers, Accessing the address of variable, Declaring and initializing pointers, accessing a variable through its pointer, Pointer expressions, Pointer increments and scale factor, Pointers and arrays, Pointers and character strings, Pointers and Functions, Pointers and Structures.	
Structures and Unions	
Introduction, Structure definition, Giving values to members, Structure initialization, Comparison of structures, Arrays of structures, Arrays within structures, Structures within Structures, Structures and functions, Unions.	
File Handling	
Introduction, Defining and opening File, Closing File, Input/output operations on Files.	

Method of delivery

Chalk and Board, Power Point presentation

Study time

3 hours Theory, 2 hours Practical

CO-PO Mapping (PO: Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	-	2	2	-	-	1	-	3
CO2	3	3	1	-	-	-	-	-	-	-	-	3
CO3	1	3	3	1	2	1	-	-	-	2	1	3
CO4	2	3	3	2	2	1	-	-	-	-	1	2
CO5	3	3	3	2	2	1	-	-	1	-	1	3
CO6	3	2	3	3	2	1	3	-	2	2	1	3

Blooms Taxonomy and Knowledge retention (For reference)

(Bloom's taxonomy has been given for reference)

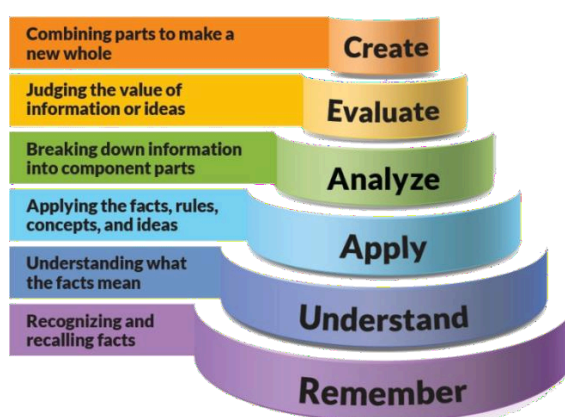


Figure 1: Blooms Taxonomy

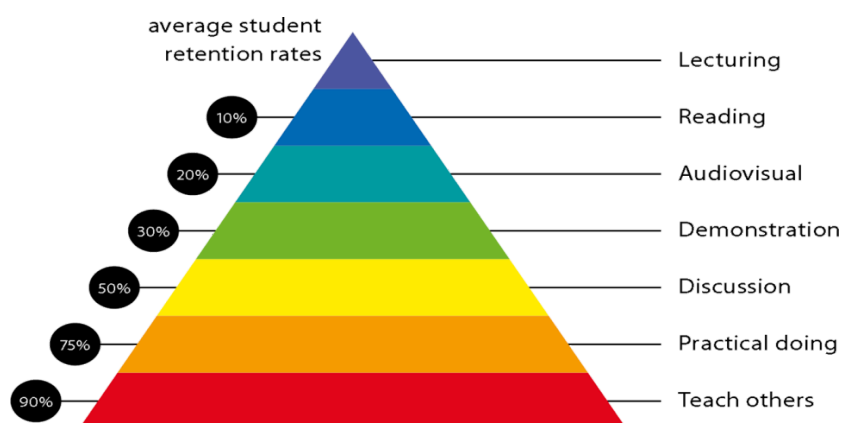


Figure 2: Knowledge retention

Graduate Qualities and Capabilities covered

(Qualities graduates harness crediting this Course)

General Graduate Qualities	Department of Computer Engineering Graduate Capabilities
Informed Have a sound knowledge of an area of study or profession and understand its current issues, locally and internationally. Know how to apply this knowledge. Understand how an area of study has developed and how it relates to other areas.	1 Professional knowledge, grounding & awareness
Independent learners Engage with new ideas and ways of thinking and critically analyze issues. Seek to extend knowledge through ongoing research, enquiry and reflection. Find and evaluate information, using a variety of sources and technologies. Acknowledge the work and ideas of others.	2 Information literacy, gathering & processing
Problem solvers Take on challenges and opportunities. Apply creative, logical and critical thinking skills to respond effectively. Make and implement decisions. Be flexible, thorough, innovative and aim for high standards.	4 Problem solving skills
Effective communicators Articulate ideas and convey them effectively using a range of media. Work collaboratively and engage with people in different settings. Recognize how culture can shape communication.	5 Written communication 6 Oral Communication 7 Teamwork
Responsible Understand how decisions can affect others and make ethically informed choices. Appreciate and respect diversity. Act with integrity as part of local, national, global and professional communities.	10 Sustainability, societal & environmental impact

Practical work:

UNIT	TOPIC/DEFINITION	HOURS												
I	Using input and output statements, Operators	4												
	1.1 Write a program to print the address of INDUS.													
	1.2 Write a program to perform basic arithmetic operators on given two numbers.													
	1.3 Find the area and perimeter of square and rectangle and circle. Input the side(s) through the keyboard. (use PI as symbolic constant)													
	1.4 Write a program to swap values of 2 variables (i) with extra variable and (ii) without using an extra variable.													
	1.5 Write a program to print the ASCII value of a given character.													
	1.6 Write a program to enter the integer number and convert it into Rs and Paisa.													
	1.7 Write a program to enter two numbers. Make the comparison between them with a conditional operator. If the first number is greater than second perform multiplication otherwise division operation.													
	1.8 Write a program to enter the temperature in Fahrenheit and convert it to Celsius. $[C = ((F-32)*5)/9]$													
	1.9 Write a program to calculate simple interest.													
	1.10 Write a program to enter marks of five subject of a student and calculate its average, percentage.													
II	Using conditional statements	4												
	2.1 Write a program to find the maximum of (i) two integer values and (ii) three integer values.													
	2.2 Write a program to check whether the given character is a vowel or not.													
	2.3 Write a program that reads a number from 1 to 7 and accordingly it should display MONDAY to SUNDAY (if- else if).													
	2.4 Write a menu driven program to perform the arithmetic operations.													
	2.5 Write a program to print the number of days in a given month using switch statements. The program requires a month number (between 1 to 12) as an input and then displays the number of days in that month.													
	2.6 Write a program to check whether a given value is even or odd.													
	2.7 Write a program to calculate total salary of an employee.													
	total salary = basic + da + hra + ta. da = 50% of basic. <table> <tr> <th>Basic</th><th>hra</th><th>ta</th></tr> <tr> <td><6000</td><td>400</td><td>100</td></tr> <tr> <td>6001>= & <10000</td><td>1400</td><td>300</td></tr> <tr> <td>>=10000</td><td>2400</td><td>700</td></tr> </table>	Basic	hra	ta	<6000	400	100	6001>= & <10000	1400	300	>=10000	2400	700	
Basic	hra	ta												
<6000	400	100												
6001>= & <10000	1400	300												
>=10000	2400	700												
III	Using control statements	5												
	3.1 Write a program to print 1 to 10 numbers using a while loop.													
	3.2 Write a program to read any 7 numbers and print the average value using for loop.													
	3.3 Write a program to reverse a given integer number.													
	3.4 Write a program to print Fibonacci series of given number.													
	3.5 Write a program to find factorial of a number.													

	3.6	Write a program to check whether a number is a Krishnamurthy number or not. Krishnamurthy number is one whose sum of factorial of digits equals the number. Example:145 $1! + 4! + 5! = 1 + 24 + 120 = 145$	
	3.7	Write a program to check whether the number is Armstrong or not. Example:153---- $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$	
	3.8	Write a program to list all prime numbers within a given range.	
	3.9	Write a program to draw following patterns: <pre> * 1 5 4 3 2 1 A ** a b 4 3 2 1 A B *** 1 2 3 3 2 1 A B C **** a b c d 2 1 A B C D ***** 1 2 3 4 5 1 A B C D E 1 1 121 01 12321 1 01 1234321 0 1 01 </pre>	
IV	Array And Strings		6
	4.1	Write a program to read 10 integers in an array. Find the addition of all elements.	
	4.2	Write a program to find the number of odd and even elements from the 1- D array.	
	4.3	Write a program to reverse the elements of an array and store it in another array.	
	4.4	Write a program to sort elements of array.	
	4.5	Write a Program to print Addition of two matrices.	
	4.6	Program to remove duplicate numbers from a list of numbers and print the list without duplicate numbers.	
	4.7	Write a Program to print Multiplication of two matrices.	
	4.8	Read the marks of five subjects obtained by five students in an examination. Display the top two student's codes and their marks.	
	4.9	Write a program to insert an element in an array at specified position.	
	4.10	Write a program to find the length of a string.(without inbuilt Function)	
	4.11	Write a program to reverse the string.(without inbuilt Function)	
	4.12	Write a program to convert a string in to lower case and upper case.	
	4.13	Write a menu driven program for the implementation of all build-in string functions.	
	4.14	Program to extract n characters starting from m in a given string. (String, n and m should be provided as inputs).	
	4.15	Find out occurrence of each character in a given string.	
V	Structure & Union		3
	5.1	Write a program to define structure with tag state with fields state name, number of districts and total population. Read and display the data.	
	5.2	Write a program to create a structure of 5 student's roll_no and name and display the records. Use array of structure	

	5.3	Write a program to create structure of bank with accno, holder_name and balance and display them for n holders whose balance is less than 5000.	
	5.4	Write a program to create union of student's roll_no and name and display the records.	
VI	Pointers & Functions		8
	6.1	Write a program that demonstrates the use of address of (&) and pointer (*) operator.	
	6.2	Write a program to read and display values of an integer array. Allocate space dynamically for the array.	
	6.3	Write a program to display the content of 1-D array using pointer.	
	6.4	Write a program to sum given two integer numbers using function.	
	6.5	Write a program using a function to count the area of circle, triangle, rectangle and square.	
	6.6	Write a program using the user defined function even_odd. With argument and check whether the no is even or odd.	
	6.7	Write a program using function with array, take input of five subject's marks and count the percentage and display result.	
	6.8	Write a function which accepts a character array as an argument from the user. The function should convert all the lowercase characters into uppercase case	
	6.9	Write a function using pointer parameter that calculate maximum element from given array of integer number.	
	6.10	Write a program that demonstrates call by value and call by reference concept in function argument.	

Lecture/ Tutorial times

Lecture and Lab as per faculty time table.

Attendance Requirements

The University norms state that it is the responsibility of students to attend all lectures, tutorials, seminars and practical work as stipulated in the course outline. Minimum attendance requirement as per university norms is compulsory for being eligible for semester examinations.

Text books

1. Programming in ANSI C by Balagurusamy, publisher: TMH

Reference Books:

1. Introduction to C by Reema Thareja, Publisher-Oxford
2. Programming with ANSI and Turbo C, by Ashok N Kamthane, Publisher – Pearson Education.
3. Let us C, by Yashwant Kanetkar, Publisher – BPB Publication

Additional Materials

Web Resource:

<http://nptel.ac.in/courses/106105085/2>

https://onlinecourses.nptel.ac.in/iitk_cs_101/previe

https://onlinecourses.nptel.ac.in/noc15_cs15/preview

ASSESSMENT GUIDELINES

Your final course mark will be calculated from the following:

CIE-Theory (60 Marks) Class Regularity – 5 Marks Mid Sem- 40 Marks Assignments/Quiz/Viva - 15 Marks	CIE-Practical (60 Marks) Practical Performance – 20 Marks Practical Test - 20 Marks Practical File & Viva - 20Marks
ESE-Theory- 40 Marks	ESE-Practical-40 Marks
Total: 200 Marks	

SUPPLEMENTARY ASSESSMENT

Students who receive an overall mark less than 40% in internal component or less than 40% in the end semester will be considered for supplementary assessment in the respective components (i.e internal component or end semester) of semester concerned. Students must make themselves available during the supplementary examination period to take up the respective components (internal component or end semester) and need to obtain the required minimum 40% marks to clear the concerned components.

Practical Work Report/Laboratory Report:

A report on the practical work is due the subsequent week after completion of the class by each group.

Late Work

Late assignments will not be accepted without supporting documentation. Late submission of the reports will result in a deduction of -% of the maximum mark per calendar day

Format

All assignments must be presented in a neat, legible format with all information sources correctly referenced. **Assignment material handed in throughout the session that is not neat and legible will not be marked and will be returned to the student.**

Retention of Written Work

Written assessment work will be retained by the Course coordinator/lecturer for two weeks after marking to be collected by the students.

University and Faculty Policies

Students should make themselves aware of the University and/or Faculty Policies regarding

plagiarism, special consideration, supplementary examinations and other educational issues and student matters.

Plagiarism - Plagiarism is not acceptable and may result in the imposition of severe penalties. Plagiarism is the use of another person's work, or idea, as if it is his or her own - if you have any doubts at all on what constitutes plagiarism, please consult your Course coordinator or lecturer. Plagiarism will be penalized severely.

Do not copy the work of other students.

Do not share your work with other students (except where required for a group activity or assessment).

Course schedule (subject to change)

(Mention quiz, assignment submission, breaks etc as well in the table under the Teaching Learning Activity Column)

Week No	Topic & Contents	CO Addressed	Teaching Learning Activity
1	Introduction to Programming: What is programming? Problem solving methods with Examples-Algorithm and Flowchart, Types of Programming languages, Characteristics of higher level language, Some Programming languages	1,2,3	Chalk & Board/Power Point Presentation/Quiz
2	Introduction to 'C': Importance of C, Sample C programs, Basic structure of C programs, Programming style, executing a C program.	2,3	Chalk & Board/Power Point Presentation
3	Introduction, Character Set, C tokens, Keywords and Identifiers, Constants, Variables, Data types, Declaration of Variables, Defining symbolic constants	2,3	Chalk & Board/Power Point Presentation
4	Operators and Expression: Arithmetic of Operators, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operators, Bitwise Operators, Special Operators, Arithmetic Expressions, Evaluation of expressions, Precedence of arithmetic operators, Type conversions in expressions, Mathematical function	3,4	Chalk & Board/Power Point Presentation/Assignment
5	Decision Making Statements: Introduction, Decision making with IF statement, Simple IF statement, the IF ELSE statement, Nesting of IF ... ELSE statements	3,4,5	Chalk & Board/Power Point Presentation
6	The ELSE IF ladder, The switch statement, the ternary (?:) Operator, the GOTO statement	3,4,5	Chalk & Board/Power Point Presentation
7	Looping: WHILE statement, the DO statement, The FOR statement, Jumps in loops Break and continue	4,5	Chalk & Board/Power Point Presentation/Program based exercises
8	Array & Handling of Character strings: Introduction, One-dimensional arrays, Two-dimensional arrays, Multidimensional Arrays	4,5	Chalk & Board/Power Point Presentation
9	Handling of Character strings: Introduction, Declaring and initializing string variables, reading string from terminal, writing string to screen,	4,5	Chalk & Board/Power Point Presentation

	Arithmetic operations on characters, Putting string together,		
10	String Operations: String Copy, String Compare, String Concatenation and String Length, String Handling functions, Table of strings	4,5	Chalk & Board/Power Point Presentation
11	User-Defined Functions: Introduction, need for user-defined functions, return values and their types, calling a function, category of functions, no arguments and no return values, Arguments with return values	4,5	Chalk & Board/Power Point Presentation/Assignment
12	Handling of non-integer functions, Nesting of functions, Recursion, Functions with arrays, The scope and Lifetime of variables in functions	4,5	Chalk & Board/Power Point Presentation
13	Pointers: Introduction, understanding pointers, Accessing the address of variable, Declaring and initializing pointers, Accessing a variable through its pointer, Pointer expressions	4,5,6	Chalk & Board/Power Point Presentation
14	Pointer increments and scale factor, Pointers and arrays, Pointers and character strings, Pointers and Functions, Pointers and structures	4,5,6	Chalk & Board/Power Point Presentation
15	Structures and Unions: Introduction, Structure definition, Giving values to members, Structure initialization, Comparison of structures, Arrays of structures, Arrays within structures, Structures within Structures, Structures and functions, Unions	4,5,6	Chalk & Board/Power Point Presentation
16	File Handling: Introduction, Defining and opening File, Closing File, Input/output operations on Files	4,5,6	Chalk & Board/Power Point Presentation

Subject Mind Mapping

